REVISITING CLASSIFIER TWO-SAMPLE TESTS

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ABSTRACT

The goal of two-sample tests is to assess whether two samples, $S_P \sim P^n$ and $S_Q \sim Q^m$, are drawn from the same distribution. Perhaps intriguingly, one relatively unexplored method to build two-sample tests is the use of binary classifiers. In particular, construct a dataset by pairing the n examples in S_P with a positive label, and by pairing the m examples in S_Q with a negative label. If the null hypothesis "$P = Q$" is true, then the classification accuracy of a binary classifier on a held-out subset of this dataset should remain near chance-level. As we will show, such *Classifier Two-Sample Tests* (C2ST) learn a suitable representation of the data on the fly, return test statistics in interpretable units, have a simple null distribution, and their predictive uncertainty allow to interpret where P and Q differ.

The goal of this paper is to establish the properties, performance, and uses of C2ST. First, we analyze their main theoretical properties. Second, we compare their performance against a variety of state-of-the-art alternatives. Third, we propose their use to evaluate the sample quality of generative models with intractable likelihoods, such as Generative Adversarial Networks (GANs). Fourth, we showcase the novel application of GANs together with C2ST for causal discovery.

1 INTRODUCTION

One of the most fundamental problems in statistics is to assess whether two samples, $S_P \sim P^n$ and $S_Q \sim Q^m$, are drawn from the same probability distribution. To this end, *two-sample tests* (Lehmann & Romano, 2006) summarize the differences between the two samples into a real-valued test *statistic*, and then use the value of such statistic to accept[1] or reject the null hypothesis "$P = Q$". The development of powerful two-sample tests is instrumental in a myriad of applications, including the evaluation and comparison of generative models. Over the last century, statisticians have nurtured a wide variety of two-sample tests. However, most of these tests are only applicable to one-dimensional examples, require the prescription of a fixed representation of the data, return test statistics in units that are difficult to interpret, or do not explain *how* the two samples under comparison differ.

Intriguingly, there exists a relatively unexplored strategy to build two-sample tests that overcome the aforementioned issues: training a binary classifier to distinguish between the examples in S_P and the examples in S_Q. Intuitively, if $P = Q$, the test accuracy of such binary classifier should remain near chance-level. Otherwise, if $P \neq Q$ and the binary classifier is able to unveil some of the distributional differences between S_P and S_Q, its test accuracy should depart from chance-level. As we will show, such *Classifier Two-Sample Tests* (C2ST) learn a suitable representation of the data on the fly, return test statistics in interpretable units, have simple asymptotic distributions, and their learned features and predictive uncertainty provide interpretation on *how* P and Q differ. In such a way, this work brings together the communities of statistical testing and representation learning.

The goal of this paper is to establish the theoretical properties and evaluate the practical uses of C2ST. To this end, our **contributions** are:

- We review the basics of two-sample tests in Section 2, as well as their common applications to measure statistical dependence and evaluate generative models.
- We analyze the attractive properties of C2ST (Section 3) including an analysis of their exact asymptotic distributions, testing power, and interpretability.

[1]For clarity, we abuse statistical language and write "accept" to mean "fail to reject".

- We evaluate C2ST on a wide variety of synthetic and real data (Section 4), and compare their performance against multiple state-of-the-art alternatives. Furthermore, we provide examples to illustrate how C2ST can interpret the differences between pairs of samples.
- In Section 5, we propose the use of classifier two-sample tests to evaluate the sample quality of generative models with intractable likelihoods, such as Generative Adversarial Networks (Goodfellow et al., 2014), also known as GANs.
- As a novel application of the synergy between C2ST and GANs, Section 6 proposes the use of these methods for causal discovery.

2 Two-Sample Testing

The goal of two-sample tests is to assess whether two samples, denoted by $S_P \sim P^n$ and $S_Q \sim Q^m$, are drawn from the same distribution (Lehmann & Romano, 2006). More specifically, two-sample tests either accept or reject the *null hypothesis*, often denoted by H_0, which stands for "$P = Q$". When rejecting H_0, we say that the two-sample test favors the *alternative hypothesis*, often denoted by H_1, which stands for "$P \neq Q$". To accept or reject H_0, two-sample tests summarize the differences between the two *samples* (sets of identically and independently distributed *examples*):

$$S_P := \{x_1, \ldots, x_n\} \sim P^n(X) \text{ and } S_Q := \{y_1, \ldots, y_m\} \sim Q^m(Y) \tag{1}$$

into a statistic $\hat{t} \in \mathbb{R}$. Without loss of generality, we assume that the two-sample test returns a small statistic when the null hypothesis "$P = Q$" is true, and a large statistic otherwise. Then, for a sufficiently small statistic, the two-sample test will accept H_0. Conversely, for a sufficiently large statistic, the two-sample test will reject H_0 in favour of H_1.

More formally, the statistician performs a two-sample test in four steps. First, decide a *significance level* $\alpha \in [0, 1]$, which is an input to the two-sample test. Second, compute the two-sample test statistic $\hat{t}$. Third, compute the *p-value* $\hat{p} = P(T \geq \hat{t}|H_0)$, the probability of the two-sample test returning a statistic as large as $\hat{t}$ when H_0 is true. Fourth, reject H_0 if $\hat{p} < \alpha$, and accept it otherwise.

Inevitably, two-sample tests can fail in two different ways. First, to make a *type-I error* is to reject the null hypothesis when it is true (a "false positive"). By the definition of p-value, the probability of making a type-I error is upper-bounded by the significance level α. Second, to make a *type-II error* is to accept the null hypothesis when it is false (a "false negative"). We denote the probability of making a type-II error by β, and refer to the quantity $\pi = 1 - \beta$ as the *power* of a test. Usually, the statistician uses domain-specific knowledge to evaluate the consequences of a type-I error, and thus prescribe an appropriate significance level α. Within the prescribed significance level α, the statistician prefers the two-sample test with maximum power π.

Among others, two-sample tests serve two other uses. First, two-sample tests can *measure statistical dependence* (Gretton et al., 2012a). In particular, testing the independence null hypothesis "the random variables X and Y are independent" is testing the two-sample null hypothesis "$P(X, Y) = P(X)P(Y)$". In practice, the two-sample test would compare the sample $S = \{(x_i, y_i)\}_{i=1}^n \sim P(X, Y)^n$ to a sample $S_\sigma = \{(x_i, y_{\sigma(i)})\}_{i=1}^n \sim (P(X)P(Y))^n$, where σ is a random permutation of the set of indices $\{1, \ldots, n\}$. This approach is consistent when considering all possible random permutations. However, since independence testing is a subset of two-sample testing, specialized independence tests may exhibit higher power for this task (Gretton et al., 2005).

Second, two-sample tests can *evaluate the sample quality of generative models* with intractable likelihoods, but tractable sampling procedures. Intuitively, a generative model produces good samples $\hat{S} = \{\hat{x}_i\}_{i=1}^n$ if these are indistinguishable from the real data $S = \{x_i\}_{i=1}^n$ that they model. Thus, the two-sample test statistic between $\hat{S}$ and S measures the fidelity of the samples $\hat{S}$ produced by the generative model. The use of two-sample tests to evaluate the sample quality of generative models include the pioneering work of Box (1980), the use of Maximum Mean Discrepancy (MMD) criterion (Bengio et al., 2013; Dziugaite et al., 2015; Lloyd & Ghahramani, 2015; Bounliphone et al., 2015; Sutherland et al., 2016), and the connections to density-ratio estimation (Kanamori et al., 2010; Wornowizki & Fried, 2016; Menon & Ong, 2016; Mohamed & Lakshminarayanan, 2016).

Over the last century, statisticians have nurtured a wide variety of two-sample tests. Classical two-sample tests include the t-test (Student, 1908), which tests for the difference in means of two samples; the Wilcoxon-Mann-Whitney test (Wilcoxon, 1945; Mann & Whitney, 1947), which tests for the difference in rank means of two samples; and the Kolmogorov-Smirnov tests (Kolmogorov, 1933; Smirnov, 1939) and their variants (Kuiper, 1962), which test for the difference in the empirical cumulative distributions of two samples. However, these classical tests are only efficient when applied to one-dimensional data. Recently, the use of kernel methods (Smola & Schölkopf, 1998) enabled the development of two-sample tests applicable to multidimensional data. Examples of these tests include the MMD test (Gretton et al., 2012a), which looks for differences in the empirical kernel mean embeddings of two samples, and the Mean Embedding test or ME (Chwialkowski et al., 2015; Jitkrittum et al., 2016), which looks for differences in the empirical kernel mean embeddings of two samples at optimized locations. However, kernel two-sample tests require the prescription of a manually-engineered representation of the data under study, and return values in units that are difficult to interpret. Finally, only the ME test provides a mechanism to interpret how P and Q differ.

Next, we discuss a simple but relatively unexplored strategy to build two-sample tests that overcome these issues: the use of binary classifiers.

3 Classifier Two-Sample Tests (C2ST)

Without loss of generality, we assume access to the two samples S_P and S_Q defined in (1), where $x_i, y_j \in \mathcal{X}$, for all $i = 1, \ldots, n$ and $j = 1, \ldots, m$, and $m = n$. To test whether the null hypothesis $H_0 : P = Q$ is true, we proceed in five steps. First, construct the dataset

$$\mathcal{D} = \{(x_i, 0)\}_{i=1}^n \cup \{(y_i, 1)\}_{i=1}^n =: \{(z_i, l_i)\}_{i=1}^{2n}.$$

Second, shuffle $\mathcal{D}$ at random, and split it into the disjoint *training* and testing subsets $\mathcal{D}_{\text{tr}}$ and $\mathcal{D}_{\text{te}}$, where $\mathcal{D} = \mathcal{D}_{\text{tr}} \cup \mathcal{D}_{\text{te}}$ and $n_{\text{te}} := |\mathcal{D}_{\text{te}}|$. Third, train a binary classifier $f : \mathcal{X} \to [0, 1]$ on $\mathcal{D}_{\text{tr}}$; in the following, we assume that $f(z_i)$ is an estimate of the conditional probability distribution $p(l_i = 1|z_i)$. Fourth, return the classification accuracy on $\mathcal{D}_{\text{te}}$:

$$\hat{t} = \frac{1}{n_{\text{te}}} \sum_{(z_i, l_i) \in \mathcal{D}_{\text{te}}} \mathbb{I}\left[\mathbb{I}\left(f(z_i) > \frac{1}{2}\right) = l_i\right] \tag{2}$$

as our *C2ST statistic*, where $\mathbb{I}$ is the indicator function. The intuition here is that if $P = Q$, the test accuracy (2) should remain near chance-level. In opposition, if $P \neq Q$ and the binary classifier unveils distributional differences between the two samples, the test classification accuracy (2) should be *greater* than chance-level. Fifth, to accept or reject the null hypothesis, compute a p-value using the null distribution of the C2ST, as discussed next.

3.1 Null and Alternative Distributions

Each term $\mathbb{I}\left[\mathbb{I}(f(z_i) > 1/2) = l_i\right]$ appearing in (2) is an independent Bernoulli(p_i) random variable, where p_i is the probability of classifying correctly the example z_i in $\mathcal{D}_{\text{te}}$.

First, under the null hypothesis $H_0 : P = Q$, the samples $S_P \sim P^n$ and $S_Q \sim Q^m$ follow the same distribution, leading to an impossible binary classification problem. In that case, $n_{\text{te}}\hat{t}$ follows a Binomial$(n_{\text{te}}, p = \frac{1}{2})$ distribution. Therefore, for large n_{te}, we can use the central limit theorem to approximate the null distribution of (2) by $\mathcal{N}(\frac{1}{2}, \frac{1}{4n_{\text{te}}})$.

Second, under the alternative hypothesis $H_1 : P \neq Q$, the statistic $n_{\text{te}}\hat{t}$ follows a Poisson Binomial distribution, since the constituent Bernoulli random variables may not be identically distributed. In the following, we will approximate such Poisson Binomial distribution by the Binomial$(n, \bar{p})$ distribution, where $\bar{p} = \frac{1}{n}\sum_{i=1}^n p_i$ (Ehm, 1991). Therefore, we can use the central limit theorem to approximate the alternative distribution of (2) by $\mathcal{N}(\bar{p}, \frac{\bar{p}(1-\bar{p})}{n_{\text{te}}})$.

3.2 Testing power

To analyze the power (probability of correctly rejecting false null hypothesis) of C2ST, we assume that the our classifier has an expected (unknown) accuracy of $H_0 : t = \frac{1}{2}$ under the null hypothesis "$P = Q$", and an expected accuracy of $H_1 : t = \frac{1}{2} + \epsilon$ under the alternative hypothesis "$P \neq Q$",